

Unifying Special Relativity and Quantum Mechanics

Lecture 28 — Coset Representatives, and the CCWZ Prescription

(Euler decomposition and the coset representative $\sigma(z)$,
the Goldstone fields as coordinates on the tangent plane to $\mathcal{M}_0$,
the transformation $gU(\pi) = U(\pi^g)h(\pi, g)$, the Maurer–Cartan form,
the vielbein d_μ and the connection E_μ , and the two-derivative CCWZ Lagrangian)

Typed version of the handwritten notes, with added explanations in blue boxes

Added explanation (not in the original notes) — How to read this document

Conventions. Exactly as in Lectures 1–27: normal text and the yellow / pink / grey boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, violet, pencil). Everything inside a blue box like this one is material I have added; it is not in the original notes.

Pagination — read this one carefully, there are two things going on.

1. **The first two sheets are a repeat.** This scan opens at sheet 71, but Lecture 27 already ran [p. 53*****]–[p. 72*****]. Sheets 71 and 72 are therefore duplicated from the end of that scan. Section 1 records their content compactly, clearly marked, so that this document stands on its own; for the full transcription see Lecture 27, Sections 13–14.
2. **The chapter ends and a new one begins mid-scan.** Sheets 73–78 complete the current chapter and carry the usual *twelve* stars: [p. 73*****]–[p. 78*****]. Then the numbering *restarts at 1* for a new chapter, *The CCWZ prescription*. That is the thirteenth restart of the notes, so those sheets are marked with *thirteen* stars: [p. 1*****]–[p. 10*****]. **The two star-counts are not a typo** — count them if in doubt. Twelve stars means sheets 71–78; thirteen stars means the new chapter.

Apart from the duplication, the scan is clean: eighteen sheets, 71, ..., 78, 1, ..., 10 consecutively, nothing missing, nothing out of order. PDF page n is sheet $70 + n$ for $n \leq 8$, and sheet $n - 8$ of the new chapter for $n \geq 9$.

It stops mid-development, as usual. The last sheet, [p. 10*****], begins expanding the leading CCWZ invariant in powers of the Goldstone fields and breaks off after the first line of the result. Section 9 records the fragment and a blue box carries the expansion through.

Three slips flagged in place.

1. On [p. 4*****] the vielbein is written $d_\mu^\alpha = \partial_\mu \pi / f_\pi - \dots$; the index α on the first term is missing (it should be $\partial_\mu \pi^\alpha / f_\pi$), as the α on the left-hand side and the expansion on [p. 10*****] both confirm.
2. On [p. 10*****] the contraction indices in the vielbein expansion drift ($f_{\rho\gamma\alpha}$ versus $f_{\beta\gamma\alpha}$ between the two brackets); the dummy labels are irrelevant and the result is unaffected.
3. On [p. 10*****] the last invariant is written $(\nabla_\mu d_\nu^\alpha)(\nabla^\mu d^{\alpha,\mu})$; the repeated μ is a slip for ν , i.e. $(\nabla_\mu d_\nu^\alpha)(\nabla^\mu d^{\alpha,\nu})$.

Notation carried over. T^a unbroken ($a = 1, \dots, \dim H$), X^α broken ($\alpha = 1, \dots, \dim G - \dim H$), $f_{ab\alpha} = 0$, and $(t_\pi^a)_{\alpha\beta} = -if_{a\alpha\beta}$ is the representation of H carried by the Goldstones. $U(\pi) = \exp(i\pi^\alpha(x)X^\alpha/f_\pi)$ is the Goldstone matrix.

Added explanation (not in the original notes) — Where we left off — recap of Lecture 27

Lecture 27 established that the vacuum manifold is the coset space, $\mathcal{M}_0 \cong G/H$, by partitioning G into left cosets and showing that every element of a coset gH maps the chosen vacuum Φ_0 to the same point. Counting parameters gave $\dim \mathcal{M}_0 = \dim G - \dim H$, which is the number of Goldstone bosons.

It then introduced the *Goldstone matrix* $U(\pi) = \exp(i\pi^\alpha X^\alpha/f_\pi)$, restricted to broken generators because the unbroken ones annihilate Φ_0 and are redundant. Writing $\Phi = U(\pi)\Phi_0$ makes the potential identically constant, since U is a group element and Φ_0 a minimum.

The transformation properties came out sharply different for the two halves of G : broken generators act by a *non-linear shift*, $\pi^\alpha \rightarrow \pi^\alpha - f_\pi \theta_\alpha$, which forbids every non-derivative term; unbroken ones act *linearly*, $\pi^\alpha \rightarrow [e^{-i\alpha_a t_\pi^a}]_{\alpha\beta} \pi^\beta$, with the Goldstones sitting in the representation R_π appearing in $\text{Ad}_G \xrightarrow{H} \text{Ad}_H \oplus R_\pi$. Finally the Higgs mode was restored, $v \rightarrow v + h$, giving the universal $V(h) = \frac{1}{2}(2\mu^2)h^2 + v\lambda h^3 + \frac{\lambda}{4}h^4$.

The scan stopped at the words “we start from a simple identity: a generic...”. **This lecture supplies that identity** — the Euler decomposition of an $SO(3)$ rotation — and then builds the general machinery it points to: coset representatives, the transformation rule $gU(\pi) = U(\pi^g)h(\pi, g)$, the Maurer–Cartan form, and finally the leading term of the effective Lagrangian.

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PART I — Completing the previous chapter [p. 71*****]–[p. 78*****]

1 The duplicated sheets [p. 71*****]–[p. 72*****]

Added explanation (not in the original notes) — These two sheets already appeared in Lecture 27

Sheets [p. 71*****] and [p. 72*****] are physically the same pages as the last two of the Lecture 27 scan, re-scanned here. Their content is transcribed in full in Lecture 27, Sections 13–14. For continuity, here is what they contain, compactly.

Sheet [p. 71***], second half.** The computation of $\Phi^T \Phi$ for $SO(N) \rightarrow SO(N-1)$, using $(X^\alpha)^T = -X^\alpha$ so that the two exponentials cancel:

$$\Phi^T \Phi = (v + h)^2, \quad V(h) = \frac{1}{2}(2\mu^2)h^2 + \frac{\lambda}{4}h^4 + v\lambda h^3,$$

with h the massive mode, $m_h^2 = 2\mu^2$. Then the $SU(N) \rightarrow SU(N-1)$ case, with hermitian generators $(X^\alpha)^\dagger = X^\alpha$ and $\Phi = \exp(i\pi^\alpha X^\alpha / f_\pi) \frac{1}{\sqrt{2}}(0, \dots, 0, v+h)^T$, giving $\Phi^\dagger \Phi = \frac{1}{2}(v+h)^2$.

Sheet [p. 72***].** The same potential $V(h)$ again for the $SU(N)$ case, with $m_h^2 = 2\mu^2$; then the opening of the section *Goldstone bosons as local coordinates on the vacuum manifold*, specialized to $SO(3) \rightarrow SO(2)$, with $SO(3)/SO(2) \cong S^2$, broken generators J^1, J^2 and unbroken J^3 , the sphere drawn with polar angles θ, φ , and the trailing line “we start from a simple identity: a generic...”.

The next sheet completes that sentence, and that is where the new material of this lecture begins.

2 The Euler decomposition of a rotation [p. 73*****]–[p. 74*****]

element $R \in SO(3)$ can be always written as

$$R = R_z(\phi) R_y(\theta) R_z(\psi)$$

The clearest proof is geometric. Consider $\hat{z} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

If we apply a generic rotation, $\hat{z}$ will be sent to a new unit vector $\hat{m}$

$$R \hat{z} = \hat{m} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}$$

generic $\hat{m}$ written in spherical coordinates

We observe that we can also get the same $\hat{m}$ by writing

$$R_z(\phi) R_y(\theta) \hat{z} = \hat{m}$$

We thus define $A = [R_z(\phi) R_y(\theta)]^{-1} R$
and compute

$$A \hat{z} = [R_z(\phi) R_y(\theta)]^{-1} R \hat{z} = [R_z(\phi) R_y(\theta)]^{-1} \hat{m} = \hat{z}$$

which implies that “ A ” must be some rotation around the $\hat{z}$ -axis by some angle, say ψ . We thus have $A = R_z(\psi)$ and, consequently,

$$A = (R_z(\phi) R_y(\theta))^{-1} R \stackrel{!}{=} R_z(\psi)$$

$\rightsquigarrow$

$$R = R_z(\phi) R_y(\theta) R_z(\psi)$$

“Euler z - y - z decomposition”

In group-theory language, we write

$$R = \underbrace{e^{-i\phi J^3} e^{-i\theta J^2}}_{\text{let us elaborate on this factor}} e^{-i\psi J^3}$$

$$\begin{aligned} e^{-i\phi J^3} e^{-i\theta J^2} &= \underbrace{e^{-i\phi J^3} e^{-i\theta J^2} e^{i\phi J^3}}_{\text{we use } C e^X C^{-1} = e^{C X C^{-1}}} e^{-i\phi J^3} = \\ &= \exp \left[-i\theta \underbrace{e^{-i\phi J^3} J^2 e^{i\phi J^3}}_{\text{we use } C e^X C^{-1} = e^{C X C^{-1}}} \right] e^{-i\phi J^3} \quad (\ast) \end{aligned}$$

$$\text{with } \begin{cases} C \equiv e^{-i\phi J^3} \\ X = -i\theta J^2 \end{cases}$$

We now use the fact that $\vec{J}$ transforms as a vector under rotations, namely

$$R(\vec{\theta})^{-1} J^i R(\vec{\theta}) = (R(\vec{\theta}))_{ik} J^k, \quad \text{with, in our case, } R(\vec{\theta}) \equiv R_z(-\phi)$$

Consequently, in our case

$$e^{-i\phi J^3} J^2 e^{i\phi J^3} = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}_{2k} J^k = -\sin \phi J^1 + \cos \phi J^2$$

Added explanation (not in the original notes)

What the identity is for. It is the concrete, $SO(3)$ version of the general statement proved abstractly in Lecture 27: *every group element factorizes as (something built only from broken generators) $\times$ (an element of H)*. Here $H = SO(2)$ is generated by J^3 , the broken generators are J^1, J^2 , and the Euler decomposition $R = R_z(\phi)R_y(\theta)R_z(\psi)$ is exactly such a factorization once the middle two factors are recombined — which is what [p. 74*****]–[p. 75*****] do.

Why the geometric proof works. A rotation is completely determined by where it sends the frame. Sending $\hat{z}$ to $\hat{m}$ uses up two parameters (the polar angles θ, ϕ of $\hat{m}$); the leftover freedom is a rotation about $\hat{z}$ itself, one parameter (ψ). Total $3 = \dim SO(3)$. **And notice the counting:** two parameters label the point on the sphere, i.e. the coset, and one labels the element of H — precisely $\dim(G/H) + \dim H = \dim G$, the formula of Lecture 27 [p. 58*****] made completely explicit.

The vector-transformation identity. $R^{-1}J^iR = R_{ik}J^k$ says that the generators of rotations rotate among themselves like the components of a vector — which is the adjoint representation of $SO(3)$, and the same statement as $\text{Ad}_{SO(3)} \cong$ the three-dimensional vector representation. This is the concrete instance of the conjugation formula $e^{-i\alpha T}Xe^{i\alpha T} = [e^{-i\alpha t_\pi}]X$ derived in general in Lecture 27 [p. 64*****]–[p. 65*****].

3 The coset representative $\sigma(z)$ [p. 75*****]–[p. 76*****]

Back to $(*)$

$$(*) = e^{-i\theta(-\sin\phi J^1 + \cos\phi J^2)} e^{-i\phi J^3}$$

and, consequently,

$$R = e^{-i(\theta\cos\phi J^2 - \theta\sin\phi J^1)} e^{-i\phi J^3} e^{-i\psi J^3}$$

$\rightsquigarrow$

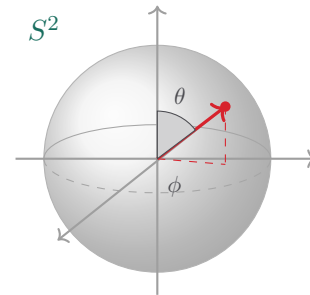
$$R = \underbrace{e^{-i(\theta\cos\phi J^2 - \theta\sin\phi J^1)}}_{\text{combination of broken generators}} \underbrace{e^{-i(\phi+\psi)J^3}}_{\text{element of the unbroken subgroup } H}$$

combination of broken generators

element of the unbroken subgroup H

We write a generic element of $G \equiv SO(3)$ as the product of the exponential of a linear combination of broken generators times an element of $SO(2)$ unbroken.

■ The two values $z \equiv (\theta, \phi)$ identify a point on the sphere S^2 , meaning that they identify an element of the coset space S^2 .



■ If we then define

$$\sigma(z) \equiv \exp\left(-i\theta\cos\phi J^2 + i\theta\sin\phi J^1\right)$$

we have a map

$$\sigma : G/H \longrightarrow G$$

that takes value on the coset space and gives group elements.

$\sigma(z)$ is called a “coset representative”: this is an element of the group that corresponds to a specific coset identified by z .

We thus find, with the help of a specific example, a general result:

$$g = \underbrace{\sigma(z)} h$$

A generic element of the group G can be always written as the product of a coset representative times an element of the unbroken subgroup.

$\sigma(z)$ is the coset representative, and it is given by the exponential of a combination of broken generators where the coefficients of such combination, identified by z , are “coordinates” on the coset space, thus on the vacuum manifold.

Added explanation (not in the original notes)

Why this is the punchline of two lectures. Lecture 27 showed abstractly that $\mathcal{M}_0 \cong G/H$ and that the Goldstone matrix is a coordinate system on it. Here that statement acquires a name and an explicit formula: $\sigma(z)$ is a *choice of one group element per coset*, and $g = \sigma(z)h$ is the factorization of an arbitrary group element into “which coset” times “where inside the coset”.

σ is a section, not a homomorphism. The map $\sigma : G/H \rightarrow G$ picks a representative from each coset, so $\pi \circ \sigma = \text{id}$ where $\pi : G \rightarrow G/H$ is the projection. It is *not* a group homomorphism — $\sigma(z_1)\sigma(z_2) \neq \sigma(z_3)$ in general, precisely because the broken generators do not close (Lecture 25 [p. 41*****]); the product of two coset representatives is another representative *times an element of H* . That leftover h is exactly the object $h(\pi, g)$ that dominates the whole of Part II.

And the choice is not unique. One could equally define $\sigma'(z) = \sigma(z)h_0(z)$ for any z -dependent $h_0 \in H$; both are legitimate coset representatives. Choosing the pure exponential of broken generators, as here, is the *canonical* choice — it is what makes the coordinates z into the Goldstone fields. Any other choice amounts to a field redefinition, which by the equivalence theorem of Lecture 25 [p. 24*****] leaves the S -matrix alone.

The check on the counting. $z = (\theta, \phi)$: two parameters, $\dim S^2 = 2$; and h carries one, $\dim SO(2) = 1$. Three altogether, matching $\dim SO(3)$.

4 The Goldstone fields as coordinates on the tangent plane

[p. 77*****]–[p. 78*****]

We thus have a geometrical interpretation of the Goldstone matrix

$$\mathcal{U}(\pi) = \exp\left(\frac{i\pi^\alpha(x)X^\alpha}{f_\pi}\right)$$

$\pi^\alpha : \mathcal{M}_{\text{spacetime}} \rightarrow \mathbb{R}$. For every fixed space-time point, $\pi^\alpha(x)$ are real numbers.

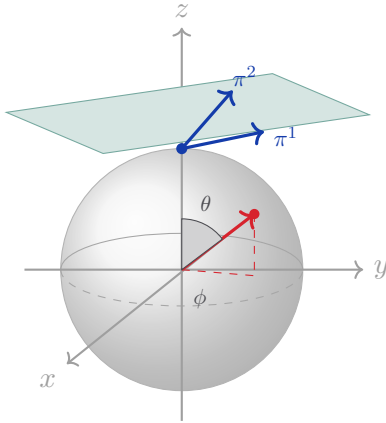
- We exponentiate a real linear combination of broken generators, so this is a coset representative.
- Consequently, $\pi^\alpha(x)$ play the role of “ z ”, and are coordinates on the coset space.

We can be more precise with the help of the previous example.

$$\begin{aligned}\sigma(z) &= \exp\left(-i\theta \cos \phi J^2 + i\theta \sin \phi J^1\right) = \\ &= \exp\left(\frac{i\pi^1 J^1 + i\pi^2 J^2}{f_\pi}\right)\end{aligned}$$

We identify
$$\begin{cases} \frac{\pi^1}{f_\pi} = \theta \sin \phi \\ \frac{\pi^2}{f_\pi} = -\theta \cos \phi \end{cases} \quad \begin{cases} \pi^1 = f_\pi \theta \sin \phi \\ \pi^2 = -f_\pi \theta \cos \phi \end{cases}$$

We thus have:



tangent plane to the vacuum manifold at the chosen vacuum.

$$\begin{cases} \pi^1 = f_\pi \theta \sin \phi \\ \pi^2 = -f_\pi \theta \cos \phi \end{cases}$$

Meaning that:

The Goldstone matrix $\mathcal{U}(\pi) = \exp(i\pi^\alpha(x)X^\alpha/f_\pi)$ is a **coset representative**

and

The Goldstone fields **locally parametrize the tangent plane to the vacuum manifold at the chosen vacuum**

Added explanation (not in the original notes)

Reading the identification. π^1 and π^2 are, up to the factor f_π , the Cartesian components of the vector of length θ pointing in the azimuthal direction ϕ — that is, $(\pi^1, \pi^2)/f_\pi = \theta(\sin \phi, -\cos \phi)$ has modulus $\sqrt{(\pi^1)^2 + (\pi^2)^2}/f_\pi = \theta$, the polar angle, and its direction encodes ϕ . So the Goldstone fields are *polar coordinates flattened out*: they are the components in the tangent plane of a vector whose length is the geodesic distance from the north pole. (The apparent mismatch of which trigonometric function goes with which index, and the minus sign, is just the fixed convention J^1, J^2 inherited from the Euler decomposition.)

Why “locally” and why “tangent plane”. Near the pole, θ small, the sphere is well approximated by its tangent plane and the map $(\pi^1, \pi^2) \mapsto \text{point of } S^2$ is a good chart. Far away it degenerates: $\theta = \pi$ (the south pole) is reached from every azimuthal direction at once, so the whole circle $\sqrt{(\pi^1)^2 + (\pi^2)^2} = \pi f_\pi$ maps to a single point. This is the exponential map of Riemannian geometry, and it is the precise reason for the word “local”. Perturbation theory only ever expands around $\pi = 0$, so this is never a problem in practice.

The physical content. A Goldstone field configuration is a map from spacetime into the

vacuum manifold: at each x , the fields say which vacuum the system is sitting in *there*. The energy comes entirely from the fact that neighbouring points may choose different vacua — hence derivatives, hence a kinetic term and nothing else. This is why Goldstone bosons are massless and derivatively coupled, stated geometrically. Everything in Part II is the systematic construction of that kinetic term.

PART II — The CCWZ prescription [p. 1*****]–[p. 10*****]

Added explanation (not in the original notes) — New chapter — the star count changes here

From this point on the handwritten page numbers restart at 1, so the margin markers carry *thirteen* stars, [p. 1*****]–[p. 10*****], rather than the twelve of [p. 78*****] and everything before it in this chapter. Nothing is missing; a new chapter simply begins.

5 The problem, and the transformation rule for $U(\pi)$ [p. 1*****]–[p. 2*****]

■ The CCWZ prescription

For a given breaking pattern $G \rightarrow H$, we would like to write an EFT describing only the Goldstone fields.

↔ Not easy since the transformation properties of the Goldstones under the broken part of G is non-linearly realized.

Instead of working with the Goldstone fields $\pi^\alpha(x)$, let us consider the Goldstone matrix

$$\mathcal{U}(\pi) = \exp\left(\frac{i\pi^\alpha(x)X^\alpha}{f_\pi}\right)$$

Let us act on $\mathcal{U}(\pi)$ with some group element $g \in G$.

$g\mathcal{U}(\pi)$ is still an element of the group. Therefore we will write it as the product of a coset representative times an element of the unbroken subgroup

$$g\mathcal{U}(\pi) = \underbrace{\mathcal{U}(\pi^g)}_{\text{coset representative}} \underbrace{h(\pi, g)}_{\text{element of the unbroken subgroup}}$$

↗ coset representative, different from $\mathcal{U}(\pi)$, that we indicate with $\mathcal{U}(\pi^g)$

↖ element of the unbroken subgroup H (that depends on π and g)

We thus get

$$\mathcal{U}(\pi^g) = g\mathcal{U}(\pi) h(\pi, g)^{-1}$$

Based on this consideration, we write the action of the group G on the Goldstone matrix as follows

$$\mathcal{U}(\pi) \xrightarrow{G} \mathcal{U}(\pi^g) = g\mathcal{U}(\pi) h(\pi, g)^{-1}$$

★

Note that, if we write $\Phi(x) = \mathcal{U}(\pi)\Phi_0$, the above transformation induces on $\Phi(x)$ the transformation

$$\Phi(x) = \mathcal{U}(\pi)\Phi_0 \xrightarrow{G} g\mathcal{U}(\pi) \underbrace{h(\pi, g)^{-1}\Phi_0}_{=\Phi_0 \text{ since } h^{-1} \in H} = g\mathcal{U}(\pi)\Phi_0 = g\Phi(x)$$

That is, the correct linear transformation realized at the level of $\Phi(x)$.

★, therefore, reflects the action of G on the Goldstone matrix.

It is, however, unclear how to use it to construct objects that are invariant under the action of G . The key idea is to consider the following object

$$\mathcal{U}(\pi(x))^{-1} \partial_\mu \mathcal{U}(\pi(x))$$

where we highlight explicitly the “ x ” dependence of $\pi(x)$.

Added explanation (not in the original notes)

The whole difficulty in one sentence. We want a Lagrangian for the π^α alone. But under the broken part of G they shift, $\pi^\alpha \rightarrow \pi^\alpha - f_\pi \theta_\alpha$ plus field-dependent corrections, so it is not obvious how to build anything invariant out of them directly. The trick is to stop working with the fields and work with the *matrix* $\mathcal{U}(\pi)$, whose transformation law, though it involves the awkward $h(\pi, g)$, is at least a clean matrix statement.

Why h depends on π as well as g . $g\mathcal{U}(\pi)$ is some group element; factorizing it into (coset representative) $\times$ (H element) is possible and unique, but *which* H element comes out depends on where you started — i.e. on π — as well as on g . This is exactly the leftover noted in Lecture 27: the product of two coset representatives is not a coset representative. For $g \in H$ one gets simply $h(\pi, g) = g$, with no π dependence, recovering the linear action of Lecture 27 [p. 65*****]. For g in the broken directions, h genuinely depends on the fields — and that field dependence is the technical content of the phrase “non-linear realization”.

The consistency check with Φ . It is worth pausing on the displayed computation. It shows the construction is not arbitrary: at the level of the original field Φ , which lives in a linear representation, the transformation is the honest linear one $\Phi \rightarrow g\Phi$. All the awkwardness has been pushed into the coordinates we chose on the vacuum manifold. Same physics, different bookkeeping — exactly the point made about field redefinitions in Lecture 25 [p. 24*****].

Why $\mathcal{U}^{-1} \partial_\mu \mathcal{U}$ is the right object to try. Under ★ the g factor is a *constant* matrix, so it cancels between $\mathcal{U}^{-1}$ and $\mathcal{U}$; and the derivative kills nothing else. What survives is a transformation involving only $h(\pi, g)$ — an element of the unbroken group, under which everything is linear. This is the manoeuvre that converts a non-linearly realized G into a linearly realized H , and it is the heart of the CCWZ construction (Callan, Coleman, Wess and Zumino, 1969).

6 The Maurer–Cartan form: vielbein and connection [p. 3*****]— [p. 4*****]

To evaluate this object, we will use the standard identity for the derivative of a matrix exponential

$$\begin{aligned} e^{-A(x)} \frac{d e^{A(x)}}{dx} &= \int_0^1 ds \, e^{-sA(x)} \frac{dA(x)}{dx} e^{sA(x)} && \text{using BCH} \\ &= A' + \frac{1}{2} [A', A] + \frac{1}{3!} [[A', A], A] + \cdots && \text{and integrating} \end{aligned}$$

We use this formula with

$$A(x) \equiv \frac{i\pi^\alpha(x)X^\alpha}{f_\pi}, \quad A' = \frac{i(\partial_\mu \pi^\alpha(x))X^\alpha}{f_\pi}$$

We thus write

$$\begin{aligned}
 \mathcal{U}(\pi)^{-1} \partial_\mu \mathcal{U}(\pi) &= \frac{i(\partial_\mu \pi^\alpha) X^\alpha}{f_\pi} + \frac{1}{2} \left[\frac{i(\partial_\mu \pi^\alpha) X^\alpha}{f_\pi}, \frac{i\pi^\beta X^\beta}{f_\pi} \right] + \dots \\
 &= \frac{i(\partial_\mu \pi^\alpha) X^\alpha}{f_\pi} + \frac{1}{2} \frac{i(\partial_\mu \pi^\alpha)}{f_\pi} \frac{i\pi^\beta}{f_\pi} \underbrace{[X^\alpha, X^\beta]}_{= i f_{\alpha\beta a} T^a + i f_{\alpha\beta\gamma} X^\gamma} + \dots \\
 &= \frac{i(\partial_\mu \pi^\alpha) X^\alpha}{f_\pi} - \frac{1}{2f_\pi^2} (\partial_\mu \pi^\alpha) \pi^\beta \left(i f_{\alpha\beta a} T^a + i f_{\alpha\beta\gamma} X^\gamma \right) + \dots
 \end{aligned}$$

$$\begin{aligned}
 \mathcal{U}(\pi)^{-1} \partial_\mu \mathcal{U}(\pi) &= \left[\frac{i(\partial_\mu \pi^\alpha)}{f_\pi} - \frac{i}{2f_\pi^2} (\partial_\mu \pi^\beta) \pi^\gamma f_{\beta\gamma\alpha} + \dots \right] X^\alpha \\
 &\quad + \left[-\frac{i f_{\alpha\beta a}}{2f_\pi^2} (\partial_\mu \pi^\alpha) \pi^\beta + \dots \right] T^a
 \end{aligned}$$

The important conceptual point here is that higher orders in the expansion will always contribute either to terms proportional to X^α or to T^a .

We thus write

Maurer–Cartan form

$$\mathcal{U}(\pi)^{-1} \partial_\mu \mathcal{U}(\pi) = i d_\mu^\alpha(\pi) X^\alpha + i E_\mu^a(\pi) T^a \equiv i d_\mu + i E_\mu$$

$$d_\mu^\alpha(\pi) = \frac{\partial_\mu \pi^\alpha}{f_\pi} - \frac{(\partial_\mu \pi^\beta) \pi^\gamma f_{\beta\gamma\alpha}}{2f_\pi^2} + \dots$$

$$E_\mu^a(\pi) \equiv -\frac{f_{\alpha\beta a}}{2f_\pi^2} (\partial_\mu \pi^\alpha) \pi^\beta + \dots$$

called “vielbein”

called “connection”

Added explanation (not in the original notes)

Slip flagged. The vielbein box in the original reads $d_\mu^\alpha = \partial_\mu \pi / f_\pi - \dots$, without the index α on the first term. It must be $\partial_\mu \pi^\alpha / f_\pi$, as both the left-hand side and the expansion on [p. 10*****] confirm. Corrected above.

Why the split into two pieces is exhaustive. $\mathcal{U}^{-1} \partial_\mu \mathcal{U}$ is an element of the Lie *algebra* $\mathfrak{g}$ (it is the derivative of a group element brought back to the identity), and $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ splits into the unbroken generators T^a and the broken ones X^α . So *any* such object decomposes uniquely into a d_μ piece and an E_μ piece — which is the green remark about higher orders. Note that this uses $[X, X] \sim T + X$, so both pieces are genuinely populated: the connection E_μ is non-zero precisely because the broken generators do not close.

Where the names come from. They are borrowed from differential geometry, and the analogy is exact rather than decorative. On a Riemannian manifold the vielbein e_μ^a converts coordinate indices into orthonormal-frame indices, and the spin connection ω_μ makes derivatives covariant under local frame rotations. Here d_μ^α converts spacetime derivatives of the coordinates π^α into components along the tangent space to G/H in an H -covariant frame, and E_μ^a is the connection for the local H rotations of that frame. The same words were used in Lecture 20 for the tetrad formalism in gravity; the mathematics is the same.

The BCH identity, in one line. Write $F(s) = e^{-sA} \frac{d}{ds} e^{sA}$. Then $F(0) = 0$ and $F'(s) = e^{-sA} A' e^{sA}$, so integrating from 0 to 1 gives the stated integral representation. Expanding $e^{-sA} A' e^{sA} = A' + s[A', A] + \frac{s^2}{2} [[A', A], A] + \dots$ and integrating $\int_0^1 s^n ds = 1/(n+1)$ produces

the series with coefficients $1, \frac{1}{2}, \frac{1}{3!}, \dots$ — exactly the boxed result.

7 Transformation of the Maurer–Cartan form [p. 5*****]– [p. 6*****]

We now investigate the transformation properties of the Maurer–Cartan form.

$$\begin{aligned}
 \mathcal{U}(\pi)^{-1} \partial_\mu \mathcal{U}(\pi) &\xrightarrow{G} \left[g \mathcal{U}(\pi) h(\pi, g)^{-1} \right]^{-1} \partial_\mu \left[g \mathcal{U}(\pi) h(\pi, g)^{-1} \right] = \\
 &= \left[h(\pi, g) \mathcal{U}(\pi)^{-1} g^{-1} \right] \left[g (\partial_\mu \mathcal{U}(\pi)) h(\pi, g)^{-1} + g \mathcal{U}(\pi) \partial_\mu h(\pi, g)^{-1} \right] = \\
 &= h(\pi, g) \mathcal{U}(\pi)^{-1} (g^{-1} g) (\partial_\mu \mathcal{U}(\pi)) h(\pi, g)^{-1} \\
 &\quad + h(\pi, g) \mathcal{U}(\pi)^{-1} (g^{-1} g) \mathcal{U}(\pi) \partial_\mu h(\pi, g)^{-1} = \\
 &= h(\pi, g) \underbrace{\left[\mathcal{U}(\pi)^{-1} \partial_\mu \mathcal{U}(\pi) \right]}_{= id_\mu + iE_\mu} h(\pi, g)^{-1} + h(\pi, g) \partial_\mu h(\pi, g)^{-1} = \\
 &= h(\pi, g) id_\mu h(\pi, g)^{-1} + h(\pi, g) iE_\mu h(\pi, g)^{-1} \\
 &\quad + \underbrace{h(\pi, g) i(-i) \partial_\mu h(\pi, g)^{-1}}_{\text{*****}} \quad \textcircled{*}
 \end{aligned}$$

This is also a Maurer–Cartan form, which is however associated to the subgroup H .

As a consequence, if we expand it on the algebra, it will only contain terms proportional to T^a ; it is, therefore, appropriate to combine it with the connection term.

$$\textcircled{*} = h(\pi, g) id_\mu h(\pi, g)^{-1} + h(\pi, g) i(E_\mu - i\partial_\mu) h(\pi, g)^{-1}$$

We thus find the key result:

$$d_\mu(\pi) \xrightarrow{G} h(\pi, g) d_\mu(\pi) h(\pi, g)^{-1}$$

$$E_\mu(\pi) \xrightarrow{G} h(\pi, g) [E_\mu(\pi) - i\partial_\mu] h(\pi, g)^{-1}$$

Let us unpack the content of these transformations.

Added explanation (not in the original notes)

This is the payoff. Under the *full* group G — including the broken part, which acts non-linearly on the π^α themselves — the vielbein d_μ transforms by a simple conjugation. All the non-linearity has been absorbed into $h(\pi, g)$, which is an element of the *unbroken* group. So d_μ behaves under all of G exactly like an object in a linear representation of H . Invariants are then easy to build: that is the content of [p. 9*****].

Where the inhomogeneous term comes from. The extra piece $h \partial_\mu h^{-1}$ arises because $h(\pi, g)$ depends on x through $\pi(x)$ — it is *spacetime-dependent*, even though g is constant. A constant h would give $\partial_\mu h^{-1} = 0$ and no inhomogeneous term. This single fact is why the connection is needed at all, and why it transforms inhomogeneously while the vielbein does

not.

Why the inhomogeneous term joins E_μ and not d_μ . $h \partial_\mu h^{-1}$ is the Maurer–Cartan form of H itself, so it lies entirely in $\mathfrak{h}$ — it is a combination of T^a only, with no X^α component. It therefore cannot contaminate the X -part, which is d_μ ; it can only shift the T -part, which is E_μ . That is exactly the violet remark in the notes, and it is what makes the clean separation of the two transformation laws possible.

Read the second box again. $E_\mu \rightarrow h[E_\mu - i\partial_\mu]h^{-1}$ is the transformation law of a *gauge field*. The notes spell this out on [p. 8*****]; it is worth anticipating, because it tells you in advance how E_μ will be allowed to appear in the Lagrangian — only through covariant derivatives and field strengths, never on its own.

8 Unpacking: vielbein and connection separately [p. 6*****]– [p. 8*****]

1) Transformations of the vielbein

$$\begin{aligned} d_\mu^\alpha(\pi) X^\alpha &\longrightarrow h(\pi, g) d_\mu^\alpha(\pi) X^\alpha h(\pi, g)^{-1} = \\ &= d_\mu^\alpha(\pi) \underbrace{e^{-i\theta_a(\pi, g)T^a} X^\alpha e^{i\theta_a(\pi, g)T^a}}_{\text{we computed already this term}} \quad \circledast \end{aligned}$$

This is an element of H , so we write it as $h(\pi, g) = \exp(-i\theta_a(\pi, g)T^a)$
 $(\exp(-i\theta_a(\pi, g)t_\pi^a))_{\beta\alpha} X^\beta$

$$\begin{aligned} \circledast &= d_\mu^\alpha(\pi) \left(\exp(-i\theta_a(\pi, g)t_\pi^a) \right)_{\beta\alpha} X^\beta = \\ &= \left[\exp(-i\theta_a(\pi, g)t_\pi^a) \right]_{\alpha\beta} d_\mu^\beta(\pi) X^\alpha \end{aligned}$$

We thus get

$$d_\mu^\alpha(\pi) \xrightarrow{G} \left[\exp(-i\theta_a(\pi, g)t_\pi^a) \right]_{\alpha\beta} d_\mu^\beta(\pi)$$

d_μ^α transform under the full group G as Goldstone bosons do under its unbroken part.

The key aspect is that, in terms of d_μ^α , the above transformation property is valid under the full group.

2) Transformation of the connection

We just write

$$E_\mu^a(\pi) T^a \xrightarrow{G} h(\pi, g) E_\mu^a(\pi) T^a h(\pi, g)^{-1} - i h(\pi, g) \partial_\mu h(\pi, g)^{-1}$$

This is a very evocative transformation law: let's go back to gauge theories. We found that gauge fields transform according to

$$\begin{aligned} g A_\mu^A T^A &\longrightarrow \mathcal{U} g A_\mu^A T^A \mathcal{U}^{-1} + i(\partial_\mu \mathcal{U}) \mathcal{U}^{-1} = \\ &= \mathcal{U} g A_\mu^A T^A \mathcal{U}^{-1} - i \mathcal{U} (\partial_\mu \mathcal{U}^{-1}) \end{aligned}$$

The structure of the transformation is the same!

$\hookrightarrow E_\mu^a(\pi)$ transform as if they were gauge fields associated with a local H -invariance.

$\hookrightarrow$ This is an important observation, since it implies that this object will enter in the Lagrangian via covariant derivatives and/or field strength tensors.

Added explanation (not in the original notes)

The logic of “we computed already this term”. The conjugation $e^{-i\theta_a T^a} X^\alpha e^{i\theta_a T^a} = [e^{-i\theta_a T^a}]_{\beta\alpha} X^\beta$ is exactly the result derived in Lecture 27 [p. 64*****]–[p. 65*****], with the constant parameters α_a there replaced by the field-dependent $\theta_a(\pi, g)$ here. Nothing new is needed — only the observation that the formula holds pointwise in x for any parameters, constant or not.

Why this is such a strong result. Compare the two statements:

π^α	d_μ^α
shifts non-linearly under broken G	rotates linearly, always
rotates linearly only under H	law valid under <i>all</i> of G
hard to build invariants from	easy: contract α indices

The price is that θ_a depends on the fields, so the “linear” rotation is by a field-dependent angle. But for building invariants that does not matter at all: any H -invariant contraction of α indices is invariant whatever the angle.

The gauge-field analogy, and its one difference. E_μ transforms exactly like a Yang–Mills field of Lecture 22, but with a crucial distinction: it is not an independent field. It is built out of the π^α , with no kinetic term of its own and no propagating degrees of freedom. The H invariance it is a connection for is not a real gauge symmetry either — it is the residual freedom of rotating the frame in the tangent space to G/H , which is a redundancy of the parametrization rather than of the physics. The right name for it is a *composite* or *induced* connection.

What this licenses. Because E_μ transforms inhomogeneously, it can never appear alone in an invariant. It may appear only in the combination $\nabla_\mu = \partial_\mu + iE_\mu$ acting on H -covariant objects, or inside the field strength $\mathcal{E}_{\mu\nu} = \partial_\mu E_\nu - \partial_\nu E_\mu + i[E_\mu, E_\nu]$. Both are used on [p. 10*****].

9 The CCWZ Lagrangian at the two-derivative level [p. 8*****]–[p. 10*****]

■ The CCWZ Lagrangian at the two-derivative level

We now have the basic ingredients to construct a Goldstone boson Lagrangian invariant under the action of the group G .

$\hookrightarrow$ We organize the Lagrangian in terms of an expansion in derivatives (we will comment at the end on the validity of this criterium).

Note that, in terms of derivatives,

$$[d_\mu^\alpha] = [\partial], \quad [E_\mu^a] = [\partial]$$

i) We write an invariant using d_μ^α as follows

$$\begin{aligned} \text{tr}(d_\mu d_\nu) &\xrightarrow{G} \text{tr}(h d_\mu h^{-1} h d_\nu h^{-1}) = \text{tr}(h d_\mu d_\nu h^{-1}) \\ &= \text{tr}(d_\mu d_\nu), \quad \underline{\text{invariant}} \end{aligned}$$

Note that we can also write

$$\begin{aligned} \text{tr}(d_\mu d_\nu) &= \text{tr}(d_\mu^\alpha X^\alpha d_\nu^\beta X^\beta) = d_\mu^\alpha d_\nu^\beta \underbrace{\text{tr}(X^\alpha X^\beta)}_{\delta_{\alpha\beta}} = \\ &= \delta_{\alpha\beta} d_\mu^\alpha d_\nu^\beta \end{aligned}$$

In the CCWZ construction, one typically normalizes the generators in such a way that $\text{tr}(X^\alpha X^\beta) = \delta_{\alpha\beta}$

$$\text{tr}(d_\mu d^\mu) = \delta_{\alpha\beta} d_\mu^\alpha d^{\beta,\mu}$$

This is the only invariant term with two derivatives that we can write if R_π is irreducible.

This is, therefore, the leading term of the EFT if we organize the latter as an expansion in derivatives.

ii) We can of course try to write additional invariants, but they will involve more than two derivatives.

Example: $[\text{tr}(d_\mu d^\mu)]^2 \rightsquigarrow$ invariant, but with 4 derivatives

Example: We can use E_μ^a , via covariant derivatives. We can define

$$\nabla_\mu d_\nu^\alpha \equiv \partial_\mu d_\nu^\alpha + i E_\mu^a (t_\pi^a)_{\alpha\beta} d_\nu^\beta$$

and construct invariants as $(\nabla_\mu d_\nu^\alpha)(\nabla^\mu d^{\alpha,\nu})$, but this term has again 4 derivatives.

We now expand the leading invariant in terms of the Goldstone fields. We use the explicit form of the vielbein.

$$\begin{aligned} \text{tr}(d_\mu d^\mu) &= \delta_{\alpha\beta} d_\mu^\alpha d^{\beta,\mu} = \delta_{\alpha\beta} \left[\frac{\partial_\mu \pi^\alpha}{f_\pi} - \frac{1}{2f_\pi^2} (\partial_\mu \pi^\rho) \pi^\gamma f_{\rho\gamma\alpha} + \dots \right] \\ &\times \left[\frac{\partial^\mu \pi^\beta}{f_\pi} - \frac{1}{2f_\pi^2} (\partial^\mu \pi^\rho) \pi^\gamma f_{\rho\gamma\beta} + \dots \right] = \\ &= \frac{1}{f_\pi^2} (\partial_\mu \pi^\alpha) (\partial^\mu \pi^\alpha) + \mathcal{O}\left(\frac{(\partial\pi)^2 \pi^2}{f_\pi^4}\right) \end{aligned}$$

[the scan ends here]

Added explanation (not in the original notes) — Completing the expansion — the leading CCWZ Lagrangian

The scan stops after the first line of the result. Here is where it lands.

Normalization. With $\text{tr}(d_\mu d^\mu) = \frac{1}{f_\pi^2} (\partial_\mu \pi^\alpha) (\partial^\mu \pi^\alpha) + \dots$, the leading Lagrangian must be

normalized so that the Goldstone kinetic term is canonical:

$$\mathcal{L}^{(2)} = \frac{f_\pi^2}{2} \text{tr}(d_\mu d^\mu) = \frac{1}{2} (\partial_\mu \pi^\alpha) (\partial^\mu \pi^\alpha) + \mathcal{O}\left(\frac{(\partial\pi)^2 \pi^2}{f_\pi^2}\right).$$

Free massless scalars at leading order, with all interactions suppressed by powers of π/f_π and carrying two derivatives — exactly the structure guessed by hand for a single Goldstone in Lecture 25 [p. 29*****], now derived for an arbitrary $G \rightarrow H$.

Why it deserves the name non-linear sigma model. Writing $\mathcal{L}^{(2)} = \frac{f_\pi^2}{2} g_{\alpha\beta}(\pi) \partial_\mu \pi^\alpha \partial^\mu \pi^\beta$ with $g_{\alpha\beta} = \delta_{\rho\sigma} d^\rho_\alpha d^\sigma_\beta$ identifies $g_{\alpha\beta}$ as the unique G -invariant metric on G/H (unique up to normalization when R_π is irreducible — which is precisely the caveat in the notes). The Lagrangian is the theory of maps from spacetime into the vacuum manifold, weighted by that metric. For $SO(3) \rightarrow SO(2)$ it is the familiar round metric on S^2 , $d\theta^2 + \sin^2 \theta d\varphi^2$, with θ, φ related to π^1, π^2 as on [p. 77*****].

On the “comment at the end” the notes promise about the derivative expansion.

The expansion parameter is $E/(4\pi f_\pi)$, or equivalently ∂/Λ with $\Lambda \sim 4\pi f_\pi$; each additional pair of derivatives costs a factor E^2/Λ^2 . So the two-derivative term dominates at low energy and the four-derivative ones ($[\text{tr} d_\mu d^\mu]^2$ and the covariant-derivative invariants above) are the first corrections. The criterion is valid precisely as long as $E \ll \Lambda$; above that the EFT must be UV completed, exactly as diagnosed for the single-Goldstone case in Lecture 25 [p. 31*****]–[p. 32*****]. This is the same logic that produced $c = \lambda/m_h^4$ there.

Why the counting of invariants is so restrictive. At two derivatives one has only d_μ to work with (E_μ cannot appear alone), the indices μ, ν must be contracted with $\eta^{\mu\nu}$, and the α indices must be contracted H -invariantly. If R_π is irreducible, Schur’s lemma says the only invariant bilinear form is $\delta_{\alpha\beta}$ up to normalization — hence exactly one term. If R_π is reducible, each irreducible piece gets its own independent coefficient, and there are several decay constants rather than one.

The historical payoff. For QCD, $G = SU(2)_L \times SU(2)_R \rightarrow H = SU(2)_V$, so $\dim G - \dim H = 6 - 3 = 3$ Goldstones, the pions, with $G/H \cong SU(2) \cong S^3$. R_π is the adjoint (isospin triplet) of $SU(2)_V$ and is irreducible, so there is exactly one two-derivative term and one constant — $f_\pi \simeq 92$ MeV, whence the notation used throughout these lectures. Every low-energy pion observable at leading order follows from that single number.

Summary of Lecture 28 — the logic in nine lines

Added explanation (not in the original notes) — Summary

1. The “simple identity” promised at the end of Lecture 27 is the Euler decomposition $R = R_z(\phi)R_y(\theta)R_z(\psi)$, proved geometrically: two parameters to move the pole to $\hat{m}$, one left over for rotations about $\hat{z}$. [p. 73*****]
2. Recombining the exponentials with $Ce^XC^{-1} = e^{CXC^{-1}}$ and the vector transformation of $\vec{J}$ rewrites it as $R = e^{-i(\theta \cos \phi J^2 - \theta \sin \phi J^1)}e^{-i(\phi + \psi)J^3}$: broken generators times an element of H . [p. 74*****]–[p. 75*****]
3. Defining $\sigma(z)$ as that first factor gives a *coset representative*, a map $G/H \rightarrow G$, and the general statement $g = \sigma(z)h$. The coordinates z are coordinates on the vacuum manifold. [p. 76*****]
4. Comparing with the Goldstone matrix identifies $\pi^1 = f_\pi \theta \sin \phi$, $\pi^2 = -f_\pi \theta \cos \phi$: the Goldstone fields *locally parametrize the tangent plane to $\mathcal{M}_0$ at the chosen vacuum*, and $\mathcal{U}(\pi)$ is a coset representative. [p. 77*****]–[p. 78*****]
5. *New chapter.* To build an EFT for the Goldstones one works with the matrix rather than the fields. Acting with $g \in G$ and refactorizing gives $g\mathcal{U}(\pi) = \mathcal{U}(\pi^g)h(\pi, g)$, with h depending on both g and the fields. At the level of $\Phi = \mathcal{U}\Phi_0$ this reduces to the honest linear $\Phi \rightarrow g\Phi$. [p. 1*****]–[p. 2*****]
6. The key object is $\mathcal{U}^{-1}\partial_\mu\mathcal{U}$, the *Maurer–Cartan form*. Expanding with the BCH-derived identity and splitting on the algebra gives $\mathcal{U}^{-1}\partial_\mu\mathcal{U} = id_\mu^\alpha X^\alpha + iE_\mu^a T^a$: the *vielbein* and the *connection*. [p. 3*****]–[p. 4*****]
7. Under G the constant g cancels and one finds $d_\mu \rightarrow h d_\mu h^{-1}$ and $E_\mu \rightarrow h[E_\mu - i\partial_\mu]h^{-1}$. The inhomogeneous piece lies in $\mathfrak{h}$, so it can only shift E_μ . [p. 5*****]–[p. 6*****]
8. Unpacked: $d_\mu^\alpha \rightarrow [e^{-i\theta_a t_a^\alpha}]_{\alpha\beta} d_\mu^\beta$ — i.e. d_μ transforms under the *full G* the way Goldstones transform under H alone. And E_μ^a transforms exactly like a gauge field of a local H invariance, so it may enter only through covariant derivatives and field strengths. [p. 6*****]–[p. 8*****]
9. Organizing by derivatives, $\text{tr}(d_\mu d^\mu) = \delta_{\alpha\beta} d_\mu^\alpha d^{\beta,\mu}$ is the unique two-derivative invariant when R_π is irreducible; everything else ($[\text{tr} d_\mu d^\mu]^2$, covariant-derivative invariants) has four. Expanding it gives $\frac{1}{f_\pi^2}(\partial_\mu \pi^\alpha)(\partial^\mu \pi^\alpha) + \dots$. [p. 9*****]–[p. 10*****]

Added explanation (not in the original notes) — Where this is going

The CCWZ construction is now essentially complete: given any breaking pattern $G \rightarrow H$, one writes down $\mathcal{U}(\pi)$, computes d_μ and E_μ , and builds the most general H -invariant Lagrangian order by order in derivatives. The leading term is fixed up to a single constant f_π (when R_π is irreducible); the corrections are suppressed by $E^2/(4\pi f_\pi)^2$. This is the general form of the effective theory whose simplest instance — one Goldstone, $\frac{1}{2}(\partial\pi)^2 + c[(\partial\pi)^2]^2$ — was written down by inspection in Lecture 25.

Two things remain to be said. The promised comment on the validity of the derivative expansion (its breakdown scale, and what UV completion means here) closes the chapter’s own logic. And then the larger thread: every Goldstone in this construction is a physical massless particle. Making the symmetry *local* changes that completely — the shift $\pi^\alpha \rightarrow \pi^\alpha - f_\pi \theta_\alpha$ means a local $\theta_\alpha(x)$ can set $\pi^\alpha(x) = 0$ everywhere, and the degrees of freedom reappear as longitudinal polarizations of gauge bosons that thereby acquire mass. That is the Higgs

mechanism, and the formalism assembled over Lectures 24–28 is exactly what is needed to state it in general.